



EXPLORE THE ADVANCED TECHNOLOGY DESIGNED INTO MURATA CHEERLEADERS

Mr. Koichi Yoshikawa, manager, corporate communications office, Murata Manufacturing Co., Ltd, speaks with EFY about the advanced components and technology powering the Murata Cheerleaders. It helps these robots to stabilise on the balls on which they move, synchronise with the other robots, sense its surrounding and communicate freely.

Q. What is Murata's aim behind developing the Murata Cheerleaders series of robots?

A. We develop these robots to showcase the possibilities of electronics to contribute to cultural development worldwide. Robots also convey how Murata products and technologies help to enrich life. Robots are attracting high interest, so they are one of the best ways for people to learn about Murata products.

Q. What enables the Murata Cheerleaders to stand upright so perfectly?

A. Murata's inverted-pendulum control technology can be seen in Murata Cheerleaders through the incorporation of three advanced gyro sensors. These sensors detect body tilt. Using this data, the technology quickly calculates the direction and speed of the next movement required to keep the robot balanced on top of the ball.

Q. What are the electronic parts used to implement this technology?

A. Data from the three gyro-sensors is used to quickly calculate the direction and speed of the next movement required to keep the robot balanced on top of the ball. Similar gyro sensors are used in digital cameras, car navigation systems, and in electronic stability control (ESC) systems that prevent cars from skidding. In fact, it is an essential automotive technology.

Q. How does one robot know how to keep formation with the remaining cheerleaders?

A. Ultrasonic microphone and infrared sensors in the robots receive ultrasound and infrared signals picked up by sensors. Position measurement technology measures current location by calculating the time lag between ultrasound and infrared signals picked up by sensors. The principle is similar as that of

calculating the distance of lightning by counting the seconds between the flash and thunder. The technology can detect the position in one centimetre unit. The body of the robot incorporates sensors to pick up these signals. Group-control technology lets you synchronize multiple robots efficiently to make the robot formations much more uniform (and thus more beautiful).

Q. How do these robots communicate with each other?

A. A 920 MHz communication module allows the robots to send acquired

About Murata Manufacturing Co., Ltd

Murata deals with the design, manufacture and supply of advanced electronic materials, and multifunctional, high-density modules. Their innovations can be found in a wide range of applications from mobile phones to home appliances, automotive applications to energy management systems as well as healthcare devices.

About Koichi Yoshikawa

Koichi Yoshikawa has played a key role in helping to develop the Murata Cheerleaders, Murata Manufacturing's collection of cute robots designed to cheer up people. As a control software programmer, Yoshikawa also contributed significantly to the development of the second-generation MURATA BOY, an award-winning bicycle-riding robot that debuted at CEATEC Japan in 2005. The robot features a number of Murata core technologies, including sensors for stability and redressing and detecting, obstacles and human movement. He now is guiding various promotional activities for MURATA Robots.

positioning information to the control system. The 920 MHz band is the same used for monitoring bridges and buildings because it is capable of long-distance communication long-term monitoring (versus Bluetooth and Wi-Fi for short-distance communication).

Q. Where can these technologies be applied in real world situations?

A. Sensing and communication technology will be applied in energy management system of houses and buildings, and to transmit people's vital signals for advanced healthcare. Group control technology could be applied in rescue robots designed to perform group tasks at disaster sites. Also, technology for the co-ordination of moving objects is expected to realize safer and more efficient vehicles and transportation systems in the future.

Q. What are the most remarkable new features of Murata Cheerleaders?

A. We have advanced this technology in collaboration with Kyoto University, a leading centre of research into intersection algorithms, group behaviour, and collision avoidance. We also paid special attention to the aesthetic aspects of the group formations, hoping to impress and even surprise audiences. Murata's collaboration with outside specialists is another one of the distinguishing features of this new robot.

Q. What is Murata's business strategy?

A. We will accelerate our development in new applications and expand our businesses in emerging markets, as well as further adding value to our products and solutions in the communication and automotive industries where Murata traditionally has strengths in.

We will also focus on new applications, such as environment, energy and healthcare in addition to automotive.